

Measuring Diversity and Segregation with Convex Functions: Supplementary Material

Philip S. Chodrow *

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1 Pielou’s Third Diversity Axiom

The last of Pielou’s three axioms relates to the behavior of a diversity measure when multiple, independent demographic attributes are considered simultaneously.

Additivity under independent attributes. *“In a multiple classification (two or more qualitative variates) where the classifications are independent, the total diversity should equal the sum of the respective diversities.”* [Pie77]. This statement can be formalized as follows, interpreting “independent” in the sense of probability distributions: suppose that, for a given population $\mathcal{P} = (\mathbf{X}, \mathbf{Y}, \boldsymbol{\mu}) \in \mathbb{P}_{ndl}$, there exist two other populations $\mathcal{P}_1 = (\mathbf{X}, \mathbf{U} \in \mathbb{R}^{n \times \ell_1}, \boldsymbol{\mu})$ and $\mathcal{P}_2 = (\mathbf{X}, \mathbf{V} \in \mathbb{R}^{n \times \ell_2}, \boldsymbol{\mu})$ such that $\ell = \ell_1 \ell_2$ and a bijection $\alpha : [\ell_1] \times [\ell_2] \rightarrow [\ell]$ such that $y_{ij} = u_{i\alpha_1(j)} v_{i\alpha_2(j)}$ for all $i \in [n]$ and $j \in [\ell]$. Then $f(\mathcal{P}) = f(\mathcal{P}_1) + f(\mathcal{P}_2)$.

This axiom, in combination with Pielou’s other two and under the mild assumption of separability and continuity, is sufficient to uniquely characterize the Shannon entropy:

Theorem 1.1. *If a function $\psi : \Delta_\ell \rightarrow \mathbb{R}$ is strictly concave, continuous¹, separable, and satisfies the ad-*

ditivity under independent attributes property, then

$$\psi(\mathbf{y}) = -c \sum_{j=1}^{\ell} y_j \log y_j \quad (1)$$

for some constant $c > 0$.

A proof is given by Chakrabarti and Chakrabarty [CC05], where additivity under independent attributes corresponds to the authors’ Axiom 4.²

2 Supplementary Proofs

Proof of Theorem 2.1. Let ϕ be strictly convex. We’ll first assume that ϕ is nonnegative. Then, for any $I \subset [\ell]$ and $\lambda = \frac{|I|}{\ell}$, we have

$$\begin{aligned} \phi\left(\frac{1}{\ell} \mathbf{e}_{[\ell]}\right) &= \phi\left(\lambda \frac{1}{|I|} \mathbf{e}_I + (1-\lambda) \frac{1}{\ell-|I|} \mathbf{e}_{[\ell] \setminus I}\right) \\ &< \lambda \phi\left(\frac{1}{|I|} \mathbf{e}_I\right) + (1-\lambda) \phi\left(\frac{1}{\ell-|I|} \mathbf{e}_{[\ell] \setminus I}\right) \\ &\leq \phi\left(\frac{1}{|I|} \mathbf{e}_I\right), \end{aligned}$$

where the first inequality is by convexity and the second is by nonnegativity.

Now, relax the requirement that ϕ is nonnegative. Since the simplex Δ_ℓ is compact (closed and bounded) and ϕ is continuous, ϕ attains a minimum on Δ_ℓ , which we denote $\phi_{\min} = \min_{\mathbf{y} \in \Delta_\ell} \phi(\mathbf{y})$. Then,

*Phil is an assistant professor of computer science at Middlebury College. His email address is pchodrow@middlebury.edu.

¹Convex functions are continuous on the interior of their domain, so this stipulation requires only that they are also continuous at the boundary.

²The authors’ statement of their result also hypothesizes nonnegativity of ϕ , but this hypothesis is not required by their proof.

let $\phi'(\mathbf{y}) = \phi(\mathbf{y}) - \phi_{\min}$, which is nonnegative and convex. The calculation above now goes through replacing ϕ with ϕ' , proving that $\phi'(\frac{1}{\ell}\mathbf{e}_{[\ell]}) \leq \phi'(\frac{1}{|\mathcal{I}|}\mathbf{e}_I)$ and therefore that $\phi(\frac{1}{\ell}\mathbf{e}_{[\ell]}) \leq \phi(\frac{1}{|\mathcal{I}|}\mathbf{e}_I)$. $\square$

Proof of Theorem 2.2. To demonstrate the uniform maximizer property, for any vector $\mathbf{y} \in \Delta_\ell$,

$$\mathbf{e}_{[\ell]}/\ell = \frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \sigma(\mathbf{y}),$$

where $\mathcal{S}$ is the set of all permutations of $[\ell]$. It follows that

$$\begin{aligned} \psi\left(\frac{1}{\ell}\mathbf{e}_{[\ell]}\right) &= \psi\left(\frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \sigma(\mathbf{y})\right) \\ &< \frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \psi(\sigma(\mathbf{y})) = \psi(\mathbf{y}), \end{aligned}$$

where the inequality is by an inductive application of strict concavity and the final equality is by symmetry. $\square$

Proof of Theorem 3.2. The proof is by direct calculation. Let $\mathcal{P}' = \text{exc}_{a,b,i \leftrightarrow j}^{(\epsilon)}(\mathcal{P})$, and let $\mathbf{y}'_i$ and $\mathbf{y}'_j$ be the demographic vectors at locations i and j in $\mathcal{P}'$. Since the global demographic vector $\bar{\mathbf{y}}$ is unchanged by the exchange, we have

$$\begin{aligned} I_\phi(\mathcal{P}) - I_\phi(\mathcal{P}') &= \sum_{i=1}^n [\mu_i \phi(\mathbf{y}_i) - \mu'_i \phi(\mathbf{y}'_i)] \\ &= \mu_i [\phi(\mathbf{y}_i) - \phi(\mathbf{y}'_i)] + \mu_j [\phi(\mathbf{y}_j) - \phi(\mathbf{y}'_j)] \text{ since } \sum_{i=1}^n \mu_i \mathbf{y}_i = \bar{\mathbf{y}}. \\ &= \mu_i \nabla \phi(\mathbf{y}_i)^\top (\mathbf{y}_i - \mathbf{y}'_i) \\ &\quad + \mu_j \nabla \phi(\mathbf{y}_j)^\top (\mathbf{y}_j - \mathbf{y}'_j) \\ &\quad + o(\|\mathbf{y}_i - \mathbf{y}'_i\|) + o(\|\mathbf{y}_j - \mathbf{y}'_j\|). \end{aligned}$$

since $\mathbf{y}_k = \mathbf{y}'_k$ for all $k \notin \{i, j\}$ and $\mu_k = \mu'_k$ for all k . Using now that $\mathbf{y}_i - \mathbf{y}'_i = \lambda_i(\tilde{\mathbf{e}}_a - \tilde{\mathbf{e}}_b)$ and $\mathbf{y}_j - \mathbf{y}'_j = \lambda_j(\tilde{\mathbf{e}}_b - \tilde{\mathbf{e}}_a)$ and ignoring the higher-order terms, we have

$$I_\phi(\mathcal{P}) - I_\phi(\mathcal{P}') \simeq \mu_i \lambda_i \nabla \phi(\mathbf{y}_i)^\top (\tilde{\mathbf{e}}_a - \tilde{\mathbf{e}}_b) + \mu_j \lambda_j \nabla \phi(\mathbf{y}_j)^\top (\tilde{\mathbf{e}}_b - \tilde{\mathbf{e}}_a).$$

By the separability assumption, we have $\nabla \phi(\mathbf{y}) = (\phi'(y_1), \dots, \phi'(y_\ell))^\top$, and so

$$\begin{aligned} I_\phi(\mathcal{P}) - I_\phi(\mathcal{P}') &\simeq \mu_i \lambda_i (\phi'(y_{ia}) - \phi'(y_{ib})) + \mu_j \lambda_j (\phi'(y_{jb}) - \phi'(y_{ja})) \\ &= \epsilon (\phi'(y_{ia}) - \phi'(y_{ib}) + \phi'(y_{jb}) - \phi'(y_{ja})) \\ &= \epsilon (\phi'(y_{ia}) - \phi'(y_{ja}) + \phi'(y_{jb}) - \phi'(y_{ib})) \\ &> 0. \end{aligned}$$

The last line follows from the assumptions that $y_{ia} < y_{ja}$ and $y_{ib} > y_{jb}$, and the fact that, if ϕ is strictly convex, then ϕ' is strictly increasing, and so $\phi'(y_{ia}) - \phi'(y_{ja}) > 0$ and $\phi'(y_{jb}) - \phi'(y_{ib}) > 0$. Choosing ϵ sufficiently small to justify the linear approximation completes the proof. $\square$

Derivation of eq. (5). We have

$$\begin{aligned} I_{\phi_{\text{eq}}}(\mathcal{P}) &= \sum_{i=1}^n \mu_i (\|\mathbf{C}(\mathbf{y}_i - \mathbf{y}_{\text{ref}})\|^2 - \|\mathbf{C}(\bar{\mathbf{y}} - \mathbf{y}_{\text{ref}})\|^2) \\ &= \sum_{i=1}^n \mu_i (\|\mathbf{C}\mathbf{y}_i\|^2 - 2\mathbf{C}\mathbf{y}_i^\top \mathbf{C}\mathbf{y}_{\text{ref}} + \|\mathbf{C}\mathbf{y}_{\text{ref}}\|^2) \\ &\quad - \sum_{i=1}^n \mu_i (\|\mathbf{C}\bar{\mathbf{y}}\|^2 + 2\mathbf{C}\bar{\mathbf{y}}^\top \mathbf{C}\mathbf{y}_{\text{ref}} - \|\mathbf{C}\mathbf{y}_{\text{ref}}\|^2) \\ &= \sum_{i=1}^n \mu_i (\|\mathbf{C}\mathbf{y}_i\|^2 - 2\mathbf{C}\mathbf{y}_i^\top \mathbf{C}\mathbf{y}_{\text{ref}} - \|\mathbf{C}\bar{\mathbf{y}}\|^2 + 2\mathbf{C}\bar{\mathbf{y}}^\top \mathbf{C}\mathbf{y}_{\text{ref}}) \\ &= \sum_{i=1}^n \mu_i (\|\mathbf{C}\mathbf{y}_i\|^2 - 2\mathbf{C}(\mathbf{y}_i - \bar{\mathbf{y}})^\top \mathbf{C}\mathbf{y}_{\text{ref}} - \|\mathbf{C}\bar{\mathbf{y}}\|^2) \\ &= \left(\sum_{i=1}^n \mu_i \|\mathbf{C}\mathbf{y}_i\|^2 \right) - \|\mathbf{C}\bar{\mathbf{y}}\|^2, \end{aligned}$$

References

- [CC05] CG Chakrabarti and Indranil Chakrabarty, *Shannon entropy: axiomatic characterization and application*, International Journal of Mathematics and Mathematical Sciences **2005** (2005), no. 17, 2847–2854.
- [Pie77] Evelyn C Pielou, *Mathematical ecology*, Vol. 385, Wiley New York, 1977.